

NANO TECHNOLOGY

NOT NANO MATERIALS

→ in b/w bulk solids → molecules

atom $\approx 0.15 \text{ nm}$

water $\approx 0.3 \text{ nm}$

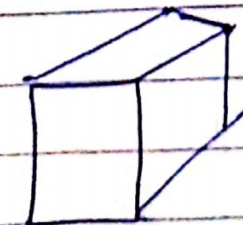
chain $\approx 80,000 \text{ nm}$

If any one dimension is reduced to less than 100 nm , that material is nanomaterial.

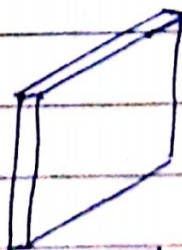
If one side is less than 100 nm
⇒ quantum well

If 2 sides are less than 100 nm
⇒ quantum wire

If all 3 sides less than 100 nm
⇒ quantum dot



Bulk



q-well



q-wire



q-dot



classification :

Fullerence .

spherical - Bucky balls .

cylindrical - C nanotubes .

Bucky tubes .

C₆₀ buckminster fullerence .

2 types of cylindrical

- single walled nano tube
- multi walled nano tube .

INORGANIC NANOPARTICLES .

- nanoclusters
- nanopowders
- nanocrystal
- nanocolloid
- nanoring
- nanorod
- nanoshell .

8) why nanoparticles show excellent pty?

- 1) Surface to volume ratio.
- 2) Quantum confinement effect

→ spherical particle with radius r .

$$S.A = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$

$$\text{Surface area to volume ratio} = \frac{3}{r}$$

⇒ most of the atoms will be on the surface of the metal

→ nanomaterials are in intermediate state b/w solid & isolated atoms. In nano region e^- begins to experience the confinement effect. i.e., their motion get limited by the physical size of the region or in which the domain it moves. The influence of becomes more pronounced & e^- becomes restricted by a potential barrier, that

must be overcome before they can move freely. The e^- becomes ~~requested~~ requested in what is called a potential well an enclosed region of -ve energy. When the dimension of a potential well is of the order of de-Broglie λ of e^- or mean free path of e^- , energy level of e^- change. e^- will remain confined to a small region of material. This effect is called quantum confinement effect.

METHODS OF SYNTHESIS (4 points)

2 types : top down & bottom up.

→ cutting bulk material into small nano-material
→ Physical method

→ joining together atoms to form materials of

nanosize.

→ chemical methods.